

Question	Answer	Marks
8(a)(i)	23: nucleon number or number of neutrons and protons	B1
	12: proton number or number of protons	B1
8(a)(ii)	both numbers correct for X: ${}_{11}^{23}\text{X}$	B1
	both numbers correct for beta-plus: ${}^0_1\beta^+$	B1
8(a)(iii)	lepton(s)	A1
8(b)	(the emitted particles) have a single (kinetic) energy or beta particles / decay have a (continuous) range of (kinetic) energies	B1